

TELCO CUSTOMER CHURN ANALYSIS

A Data Analytics Case Study

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Tool Stack	Excel SQL (BigQuery) Tableau R
Dataset	IBM Telco Customer Churn Kaggle.com
Dashboard	public.tableau.com/app/profile/daniel.mandisen/viz/TelcoCustomerChurnAnalysisProject
Status	Complete

PHASE 1 ASK

Business Task

Analyze customer churn patterns for a telecom company to identify the key factors that drive cancellations, determine which customer segments are at highest risk, and provide data-driven retention recommendations.

Guiding Questions

Client	Mid-size telecom company experiencing high customer cancellation rates
Stakeholders	Executive leadership, marketing team, customer retention team
Audience	Business executives and retention managers
Goal	Understand why customers are leaving and which segments are most at risk

Key Metrics

- Overall churn rate
- Churn rate by contract type
- Churn rate by customer tenure
- Churn rate by monthly charges
- Top churn reasons by frequency

PHASE 2 PREPARE

Data Sources

Dataset	IBM Telco Customer Churn
Source	Kaggle.com
License	Public free to use
Format	Excel (.xlsx)
Total Records	7,043 customers
Total Variables	33 columns
Churn Format	Yes / No (Churn Label column)
Churn Reason	Populated for 1,870 churned customers only

ROCCC Assessment

- **Reliable IBM-published dataset, widely used across the industry**
- Original Primary customer-level data
- Comprehensive 33 variables covering demographics, services, and billing
- Current Standard benchmark dataset used in active data science practice
- Cited Source fully documented above

Potential Limitations

- Dataset is simulated and anonymized not sourced from a real company
- Churn Reason column is only available for customers who churned; active customers have blank values, which is expected and not a data error

PHASE 3 PROCESS

Data Cleaning & Preparation

Step 1 Raw Data Preservation

A working copy of the raw dataset was created before any cleaning began. The original sheet was preserved as 'raw_data' and all cleaning was performed exclusively on the 'working_data' sheet. This ensures data integrity and allows comparison between raw and cleaned versions at any point in the analysis.

Step 2 Duplicate Check

Excel's Remove Duplicates function was applied across all 33 columns with headers enabled. Result: No duplicate values found. All 7,043 rows were confirmed as unique customer records.

Step 3 Blank Cell Check

Excel's Find function was used to identify all blank cells across the dataset. Result: 1,130 blank cells found, all located exclusively in the Churn Reason column. This is expected and valid only the 1,870 customers who churned have a churn reason recorded. The remaining 5,173 active customers have no churn reason by definition. These blanks were left as-is and are not treated as errors.

Step 4 Data Type Verification

CustomerID	Alphanumeric text confirmed correct
Tenure Months	Numerical confirmed correct
Monthly Charges	Numerical confirmed correct
Total Charges	Numerical confirmed correct
Churn Label	Text (Yes/No) confirmed correct

Step 5 Churn Columns Discovery

Four churn-related columns were identified in the dataset, providing multi-dimensional analysis capability:

- Churn Label Yes/No categorical (primary filter)
- Churn Value 0/1 numeric (used for calculations)
- Churn Score Probability score 0-100 (used for risk segmentation)
- Churn Reason Text explanation (used for qualitative insight)

Step 6 Text Formatting Consistency

Churn Label, Contract, and Gender columns were checked for capitalization consistency using Excel Find with Match Case enabled. All text values follow consistent Title Case formatting first letter capitalized, remainder lowercase. No inconsistencies or standardization needed.

Step 7 Outlier Analysis

Tenure Months	Min: 0 months Max: 72 months Normal range
Monthly Charges	Min: \$18.25 Max: \$118.75 Normal range
Total Charges	Min: \$18.80 Max: \$8,684.80 Normal range

All values fall within realistic and expected ranges for a telecom company. No outliers, negative values, or suspicious entries were identified. Numerical data confirmed clean and ready for analysis.

PHASE 4 ANALYZE

SQL Analysis BigQuery

All analysis was conducted in Google BigQuery using SQL. Seven queries were written to systematically investigate churn patterns from broad overview to granular risk profiling.

Query 1 Overall Churn Rate

Finding: 26.54% of customers (1,869 out of 7,043) have churned while 73.46% (5,174) remain active. This means roughly 1 in 4 customers is leaving the company, sitting at the high end of the telecom industry average of 20-25%.

Recommendation: Establish urgency. This churn rate represents a significant revenue loss problem that requires immediate strategic action.

Query 2 Churn by Contract Type

Finding: Month-to-month customers churn at 42.71% (1,655/3,875), compared to 11.27% for one-year contracts and just 2.83% for two-year contracts. Customers with no long-term commitment are 15x more likely to leave than two-year contract holders.

Recommendation: Incentivize customers to upgrade from month-to-month to annual contracts through targeted discounts or loyalty rewards.

Query 3 Churn by Tenure Group

Finding: New customers (0-12 months) churn at 47.44%, nearly double established customers (25-48 months) at 20.39%, and five times higher than loyal customers (49+ months) at 9.51%. The longer a customer stays, the less likely they are to leave.

Recommendation: Implement a dedicated onboarding and early retention program targeting customers in their first 12 months, including check-in touchpoints, loyalty incentives, and proactive customer service outreach.

Query 4 Churn by Monthly Charges

Finding: The high pricing tier (\$66-\$85) shows the highest churn rate at 35.89%, slightly above the very high tier (\$86+) at 33.82%. Customers in the high tier appear to feel overcharged without receiving premium value, while very high tier customers may be on bundled packages that justify the cost.

Recommendation: Review the value proposition of the \$66-\$85 pricing tier either add services, create loyalty discounts, or restructure pricing to reduce perceived price-value mismatch.

Query 5 Top 10 Churn Reasons

Finding: Three dominant themes drive departure: (1) Competition customers cited competitors offering better speeds (10.11%), more data (8.67%), better offers (7.49%), and better devices (6.96%), totaling ~43% of known churn. (2) Poor service attitude support person attitude (10.27%) and service provider attitude (7.22%) combined. (3) Product/price issues network reliability (5.51%), product dissatisfaction (5.46%), and high pricing (5.24%).

Recommendation: Address customer service training immediately as the single biggest individual churn driver, while simultaneously developing a competitive response strategy for speed, data, and devices.

Query 6 High Risk Customer Profile

Finding: The highest churn rate of 74.92% belongs to month-to-month customers in their first 12 months paying over \$86/month. The second highest at 64.04% shares the same profile but pays \$66-\$85 monthly. All top 10 highest risk combinations exclusively involve month-to-month contracts.

Recommendation: Implement an automated early intervention system that flags new month-to-month customers paying above \$66/month and triggers immediate retention action within the first 90 days of service.

PHASE 5 SHARE

Visualizations & Dashboard

An interactive Tableau dashboard was built to communicate all key findings visually. The dashboard contains five charts assembled into a single cohesive view using a consistent Red-Gold color scale where darker red indicates higher churn risk.

Dashboard Charts

- Churn Rate by Contract Type Horizontal bar chart showing 3 contract categories

- High Risk Customer Profile Heatmap combining contract, tenure, and charges
- Top 10 Reasons Customers Leave Horizontal bar chart of churn reasons
- Churn Rate by Monthly Charges Vertical bar chart by pricing tier
- Churn Rate by Tenure Group Vertical bar chart by customer tenure stage

Dashboard Link

[View Dashboard on Tableau Public](#)

PHASE 6 ACT

Top High-Level Insights

► **Insight 1 The Churn Crisis is Real**
26.54% of customers are leaving that is 1 in 4. For a telecom company with thousands of customers this represents millions in lost annual revenue. This is not a minor issue; it is a business emergency requiring immediate strategic action.

► **Insight 2 Contract Type is the #1 Lever**
Month-to-month customers churn at 42.71% compared to just 2.83% for two-year contract holders. This 15x difference means that converting even a small percentage of month-to-month customers to annual contracts would dramatically reduce overall churn.

► **Insight 3 The First Year is Make or Break**
Nearly half of new customers (47.44%) leave within their first 12 months. The company is failing to build loyalty during the most critical window of the customer relationship.

► **Insight 4 The High Risk Profile is Identifiable**
A new month-to-month customer paying over \$66/month has a 64-75% chance of churning. The company can identify these customers the moment they sign up and intervene immediately.

► **Insight 5 Customer Service is Failing**
The single biggest individual churn reason is the attitude of support staff (10.27%). This is entirely within the company's control and fixable immediately with proper training and accountability measures.

► **Insight 6 Competition is Winning on Product**

Combined competitor-related reasons account for approximately 43% of churn. Customers are leaving for better speeds, more data, and better devices revealing a product gap problem alongside the service problem.

Top Recommendations

01

Launch an Annual Contract Incentive Program

Offer month-to-month customers a meaningful discount (10-15% off) to switch to a one or two-year contract. Even converting 20% of month-to-month customers to annual contracts could reduce overall churn by 8-10 percentage points.

02

Build a First 90 Days Retention Program

Create a dedicated onboarding experience for new customers including a welcome call, a check-in at 30 days, and a loyalty reward at 90 days. Focus especially on customers paying over \$66/month on month-to-month contracts the highest risk segment.

03

Invest in Customer Service Training Immediately

Since attitude of support staff is the #1 individual churn reason, implement mandatory customer service training focused on empathy, resolution speed, and satisfaction scoring. Track improvement monthly with KPIs tied to churn outcomes.

04

Develop a Competitive Response Strategy

Audit competitor offerings on speed, data allowances, and device options. Identify where the biggest gaps are and either match competitors or create a value-add bundle that justifies current pricing in the \$66-\$85 tier.

05

Implement a Churn Prediction Alert System

Use the Churn Score column already in the dataset to flag high-risk customers automatically. Any customer with a churn score above 70 should trigger automatic retention outreach from the customer success team.

Additional Deliverables for Further Exploration

- Churn analysis by geographic region to identify location-based patterns
- Customer lifetime value calculation per segment to quantify revenue at risk
- A/B test results of retention offers to measure intervention effectiveness
- Month-over-month churn trend analysis to track improvement over time
- Churn prediction model using machine learning for proactive identification
- Sentiment analysis of customer service interactions correlated with churn